TRUNCATIONS OF THE RING OF NUMBER-THEORETIC FUNCTIONS, REVISITED

JAN SNELLMAN

Matematiska Institutionen, Linköpings Universitet, 581 83 Linköping, Sweden
jan.snellman@liu.se

Abstract. Let Γ be the ring of all functions $\mathbb{N}^+ \to K$ under Dirichlet convolution, and let Γ_n be its truncation to functions supported on $[1, n]$. In [9] it was shown that Γ_n is a polynomial ring modulo a monomial ideal I_n which is stable after reversing the order of the variables, and the Poincaré–Betti series of Γ_n was computed in terms of the numbers $C_{n,v}$ of minimal generators of I_n of least support v.

We prove that $C_{n,v} = \Phi(n, p_v)$, Legendre's sifting function: the number of integers in $[1, n]$ free of prime factors $\leq p_v$. This identifies an invariant of a minimal free resolution with a classical object of sieve theory. As consequences we obtain: a proof of Conjecture 4.6 of [9], which was left open there; the average order $C_n \sim \pi(n)^2/2$ of the total number of minimal generators, showing that the lower bound $C_n \geq \binom{\pi(n)+1}{2}$ of [9] is asymptotically sharp; and the identification of C_n with the OEIS sequence A182843. We also record errata for [9]: one stated result is false, and two proofs are incomplete. Corrected statements and complete proofs are given.

1. Introduction

Cashwell and Everett [2] studied the ring

$$\Gamma = \{ f \mid f : \mathbb{N}^+ \to K \} \tag{1}$$

of *number-theoretic functions*, with pointwise addition and the Dirichlet convolution

$$fg(n) = \sum_{ab=n} f(a)g(b); \tag{2}$$

here K is a field containing $\mathbb{Q}$. Reading the fundamental theorem of arithmetic as an isomorphism of monoids

$$(\mathbb{N}^+, \cdot) \;\cong\; \coprod_p (\mathbb{N}, +) \tag{3}$$

identifies Γ with the large power series ring $K[[x_1, x_2, \ldots]]$ on countably many variables, x_i corresponding to the i-th prime p_i; Cashwell and Everett proved that this ring is a unique factorisation domain.

In [9] the *truncations*

$$\Gamma_n = \{ f \in \Gamma \mid f(m) = 0 \text{ for all } m > n \} \tag{4}$$

were studied. Writing $r = \pi(n)$ and giving the monomial $x_1^{a_1} \cdots x_r^{a_r}$ the *weight* $\prod_i p_i^{a_i}$, so that monomials correspond bijectively to positive integers, one has

$$\Gamma_n \cong S/I_n, \qquad S = K[x_1, \ldots, x_r], \tag{5}$$

with I_n generated by all monomials of weight $> n$. The ideal I_n is strongly stable with respect to the *reversed* order of the variables, because $w(mx_j/x_i) = w(m)p_j/p_i > w(m)$ whenever $j \geq i$, that is

$$w(m\, x_j/x_i) = w(m)\,\frac{p_j}{p_i} > w(m); \tag{6}$$

this single inequality puts the Eliahou–Kervaire resolution [3] and the Golod property [1, 7, 5, 6] within reach, and yields a rational Poincaré–Betti series for Γ_n expressed in terms of the numbers

$$C_{n,v} = \#\{\, m \in G(I_n) \mid \min(m) = v\,\}, \qquad C_n = \#G(I_n), \tag{7}$$

where $G(I_n)$ is the minimal monomial generating set and $\min(m)$ is the least index occurring in m.

The purpose of this note is threefold.

First, *errata.* Theorem 3.5(3) of [9] is false, and the proofs of Theorem 3.4(4) and Theorem 3.4(5) there are incomplete. Section 4 records this and supplies corrected statements and complete proofs. The false statement is used nowhere else in [9], so nothing else in that paper is affected.

Second, and principally, a *sieve-theoretic identification.* Recall Legendre's sifting function

$$\Phi(x, y) = \#\{\, m \leq x \mid p \mid m \implies p > y\,\}, \tag{8}$$

the number of integers up to x free of prime factors $\leq y$. Our main result, Theorem 4, is that

$$\boxed{C_{n,v} = \Phi(n, p_v).} \tag{9}$$

Thus the generator counts governing a minimal free resolution over Γ_n are literally values of the classical sifting function, and the numerical tables of [9] are tables of Φ. Theorem 3.2 of [9] describes $C_{n,v}$ as the cardinality of a *different* set — the p_v-rough numbers in the interval $(n/p_v, n]$ — and Theorem 4 may be read as saying that this set is equinumerous with the much more familiar one in (8).

Third, the *consequences.* Section 5 proves Conjecture 4.6 of [9], which asserted a precise shape for the reduced Poincaré–Betti series and was left open there with numerical evidence for $n \leq 25$. Section 6 determines the average order of C_n, showing that $C_n \sim \frac{1}{2}\pi(n)^2 \sim n^2/(2\log^2 n)$ and hence that the lower bound $C_n \geq \binom{\pi(n)+1}{2}$ of Theorem 3.4(3) of [9] is asymptotically an equality. Section 8 collects what remains open.

Remark 1. The Golod criterion of [1, 7] requires the ideal to be contained in $(x_1, \ldots, x_r)^2$, and I_n is: a degree-one generator would be some x_i with $p_i > n$, and no such variable occurs in $S = K[x_1, \ldots, x_r]$ since $r = \pi(n)$. Equivalently, this is Theorem 3.5(1) of [9]. We record it because it is exactly the sort of hypothesis that goes unchecked.

Remark 2. All numerical claims below were verified by computer, along a route independent of the formulas being tested: the minimal generators of I_n were enumerated directly and their least supports tabulated, rather than computed from any of the counting formulas. The code and its raw output are available in the repository [26].

2. Notation and the counting theorem

Throughout, $p_1 = 2 < p_2 = 3 < \cdots$ are the primes, $\pi(n)$ is the number of primes $\leq n$, and $\operatorname{lpf}(m)$ denotes the least prime factor of $m > 1$. We abbreviate $r = \pi(n)$ when n is fixed, this being the number of variables of S. ([9] writes $r(n)$ throughout for what we write $\pi(n)$; we have changed the notation because the results below sit alongside the sieve-theoretic literature, where π is universal. Quotations from [9] below keep its own notation.) For a monomial $m = x_1^{a_1} \cdots x_r^{a_r}$ we write $\min(m)$ and $\max(m)$ for the least and greatest i with $a_i > 0$, and $|m| = \sum_i a_i$. We write $\lambda = \Omega$ for the number of prime factors counted with multiplicity, as in [9].

We shall use the following description, which is Theorem 3.2 of [9].

Theorem 3 ([9], Thm. 3.2)**.** *For $1 \leq v \leq \pi(n)$,*

$$C_{n,v} = \#\{\, x \in \mathbb{N} \mid n/p_v < x \leq n \text{ and } p \mid x \implies p \geq p_v \,\}.$$

It will be convenient to name the counting function appearing implicitly there. For a prime p and real $y \geq 0$ put

$$R_p(y) = \#\{\, m \leq y \mid q \mid m \implies q \geq p \,\}, \tag{10}$$

the number of p-rough integers up to y; the integer 1 is counted, vacuously. Theorem 3 then reads

$$C_{n,v} = R_{p_v}(n) - R_{p_v}(n/p_v). \tag{11}$$

Note that

$$R_p(y) = \Phi(y, p) + \#\{\, m \leq y \mid \operatorname{lpf}(m) = p \,\}, \tag{12}$$

so R_p and Φ differ precisely by the multiples of p that are p-rough.

3. The main theorem

Theorem 4. *For all $n \geq 2$ and $1 \leq v \leq \pi(n)$,*

$$C_{n,v} = \Phi(n, p_v),$$

the number of integers in $[1, n]$ having no prime factor $\leq p_v$.

Proof. Write $P = p_v$. Every P-rough integer m factors uniquely as $m = P^a m'$ with $a \geq 0$ and m' free of prime factors $\leq P$; conversely every such pair (a, m') gives a P-rough integer. Counting those with $m \leq y$ according to a gives

$$R_P(y) = \sum_{a \geq 0} \Phi(y/P^a,\ P)\,, \tag{13}$$

a finite sum, since the terms vanish once $P^a > y$. Substituting (13) into (11), the two sums telescope:

$$C_{n,v} = R_P(n) - R_P(n/P) = \sum_{a \geq 0} \Phi\big(\tfrac{n}{P^a}, P\big) - \sum_{a \geq 0} \Phi\big(\tfrac{n}{P^{a+1}}, P\big) = \Phi(n, P). \qquad \square$$

Example 5. Take $n = 25$, so $r = 9$. The integers in $[1, 25]$ with no prime factor ≤ 2 are the thirteen odd numbers, so $C_{25,1} = 13$; those with no prime factor ≤ 3 are $1, 5, 7, 11, 13, 17, 19, 23, 25$, so $C_{25,2} = 9$; those with no prime factor ≤ 5 are $1, 7, 11, 13, 17, 19, 23$, so $C_{25,3} = 7$. These agree with Figure 1 of [9]. By contrast the set of Theorem 3 for $v = 2$ is $\{9, 11, 13, 15, 17, 19, 21, 23, 25\}$ — a different set of the same size.

Splitting $[1, n]$ into 1, the primes and the composites gives at once the following reformulation, which is the form we shall use.

Corollary 6. *For* $1 \leq v \leq \pi(n)$,

$$C_{n,v} = \big(\pi(n) - v + 1\big) + E(n, v), \tag{14}$$

where

$$E(n, v) = \#\{\, x \leq n \mid x \text{ composite},\ p \mid x \implies p > p_v \,\}. \tag{15}$$

In particular

$$C_{n,v} \ \geq\ \pi(n) - v + 1,$$

with equality if and only if $p_{v+1}^2 > n$.

Proof. The integers counted by $\Phi(n, p_v)$ are 1, the primes q with $p_v < q \leq n$ — of which there are $\pi(n) - v$ — and the composites counted by $E(n, v)$. For the last sentence, $E(n, v) = 0$ exactly when there is no composite $\leq n$ all of whose prime factors exceed p_v, and the least such composite is p_{v+1}^2. $\square$

Corollary 6 contains Theorem 3.4(2) of [9] (put $v = 1 + \pi(n) - u$) and sharpens it to an exact criterion for equality. Summing over v gives a closed form for the total.

Corollary 7. $C_n = \dbinom{\pi(n) + 1}{2} + \displaystyle\sum_{\substack{x \leq n \\ x \text{ composite}}} \big(\pi(\operatorname{lpf} x) - 1\big).$

Proof. Sum Corollary 6 over $v = 1, \ldots, r$. The first terms contribute $\sum_{v=1}^{r}(r - v + 1) = \binom{r+1}{2}$. For the second, exchange the order of summation: a composite $x \leq n$ is counted by $E(n, v)$ for exactly those $v \geq 1$ with $p_v < \operatorname{lpf} x$, and there are $\pi(\operatorname{lpf} x) - 1$ of those. $\square$

Remark 8. Corollary 7 proves the increment formula

$$C_n - C_{n-1} = \begin{cases} \pi(n) & \text{if } n \text{ is prime,} \\ \pi(\operatorname{lpf} n) - 1 & \text{if } n \text{ is composite,} \end{cases}$$

recorded empirically by Costello for the OEIS sequence A182843; see Section 6. The two cases are one formula, since $\operatorname{lpf} n = n$ for n prime: the passage from $n - 1$ to n adds $\pi(n)$ to $\binom{\pi(n)+1}{2}$ when n is prime, and adds $\pi(\operatorname{lpf} n) - 1$ to the sum over composites otherwise.

4. Errata for [9]

4.1. **A false statement.** Theorem 3.5(3) of [9] asserts

$$\binom{\pi(n)}{2} = \#\{\, m \in \mathbb{N}^+ \mid m \leq n,\ \lambda(m) = 2 \,\}, \tag{16}$$

and the proof there reads, in full, "The first and the last assertions are obvious." Statement (16) is false. Its left-hand side counts *pairs of distinct primes* $\leq n$, its right-hand side counts *products of two primes that are* $\leq n$; the two agree only for $n = 2$ and $n = 4$. At $n = 10$ the left side is $\binom{4}{2} = 6$ while the right side counts $\{4, 6, 9, 10\}$, so is 4. Asymptotically the left side is $\sim n^2/(2\log^2 n)$ and the right side $\sim n \log\log n/\log n$.

The intended statement was presumably the following, which is what the proof of Theorem 3.4(2)–(3) of [9] actually establishes.

Proposition 9. *For every* $n \geq 2$,

$$\#\{\, m \in G(I_n) \mid |\mathrm{supp}(m)| = 2 \,\} \;\geq\; \binom{\pi(n)}{2},$$

with equality exactly for $2 \leq n \leq 8$.

Proof. Let $1 \leq i < j \leq r$. Choose $b \geq 1$ with $p_i^{b-1} \leq n < p_i^b$. Then

$$w(x_i^{b-1}x_j) = p_i^{b-1}p_j > p_i^{b-1}p_i = p_i^b > n,$$

so $x_i^{b-1}x_j \in I_n$ and is therefore divisible by some $m \in G(I_n)$, necessarily with $\mathrm{supp}(m) \subseteq \{i, j\}$. Now $x_i^{b-1} \notin I_n$ because $p_i^{b-1} \leq n$, so m is not a power of x_i; and $x_j \notin I_n$ because $p_j \leq n$, so m is not a power of x_j. Hence $\mathrm{supp}(m) = \{i, j\}$. Distinct pairs give distinct generators, whence the inequality. The equality range is a finite check. □

4.2. **Two incomplete proofs.** *Theorem 3.4(4) of* [9] asserts that for each v one has $C_{n,1+\pi(n)-v} = v$ for all sufficiently large n. The proof given there reads "the only integers $x \leq n$ with all prime factors $\geq 1 + r(n) - v$ are $p_{1+r(n)-v}, \ldots, p_{r(n)}$... and they are all $> n/p_v$". Two subscripts are wrong: the condition on prime factors should read $\geq p_{1+r(n)-v}$ (a prime, not an index), and the final bound should read $> n/p_{1+r(n)-v}$. With Corollary 6 the statement becomes immediate, and effective.

Proposition 10. *Fix* $v \geq 1$. *Then* $C_{n,1+\pi(n)-v} = v$ *for every* $n > 4^{v-1}$.

Proof. Put $u = 1+\pi(n)-v$. By Corollary 6, $C_{n,u} = v$ if and only if $p_{u+1}^2 > n$, where $p_{u+1} = p_{\pi(n)-v+2}$. By Bertrand's postulate there is a prime in $(y, 2y)$ for every $y \geq 1$; applied repeatedly this gives $p_{\pi(n)-j+1} > n/2^j$ for $j \geq 1$, so $p_{\pi(n)-v+2} > n/2^{v-1}$ and hence $p_{u+1}^2 > n^2/4^{v-1}$, which exceeds n as soon as $n > 4^{v-1}$. □

Theorem 3.4(5) of [9] asserts that $C_{n,v} = C_{n-1,v}$ for all v when n is even. The proof there establishes only one of the two inclusions needed. It shows that an x counted for n is counted for $n-1$; the converse, that no x is counted for $n-1$ but not for n, is not addressed. The statement is nonetheless true, and with Theorem 4 the whole matter is a triviality.

Proposition 11. *If* $n \geq 4$ *is even then* $C_{n,v} = C_{n-1,v}$ *for all* $v \geq 1$.

Proof. By Theorem 4, $C_{n,v} = \Phi(n, p_v)$ and $C_{n-1,v} = \Phi(n-1, p_v)$. These differ by 1 or 0 according as n is or is not counted by $\Phi(\cdot, p_v)$; and n is even, so it is divisible by $p_1 = 2 \leq p_v$ and is never counted. □

Remark 12. For completeness we note that the argument omitted in [9] can also be supplied directly: an integer lying in $((n-1)/p_v,\, n/p_v]$ is unique if it exists, since that interval has length $1/p_v < 1$, and it must satisfy $p_v x = n$; as n is even and p_v odd for $v > 1$, such an x is even and so is excluded by the condition on prime factors.

4.3. **Smaller items.** For the record: in equation (15) of [9] the range should be $1 \le j \le r$, not $1 \le j \le n$, since j indexes the variables $x_1, \dots, x_r$; in Theorem 3.2 and equation (19) the solution vector is written $(b_1, \dots, b_r) \in \mathbb{N}^r$ although only $b_v, \dots, b_r$ are constrained, so that as literally stated the solution set is infinite for $v > 1$ — the proof supplies the missing condition $b_j = 0$ for $j < v$ immediately, and the corresponding statement in Theorem 1.2(III) is correct as printed; the caption of Figure 2 names the quantities $C_{n,i,g}$ where the text defines $C_{n,v,d}$; and Theorem 4.3 is attributed to "Herzog–Aramova" in the text but "Aramova–Herzog" in the abstract and bibliography. Finally, we take the opportunity to note a small looseness of attribution. Theorem 4.3 of [9] is stated for *stable* ideals and credited jointly to [1] and [7]. The result of [7] (Corollary 1.2 there) assumes the ideal to be 0-Borel fixed, that is *strongly* stable, and it is in that form that the joint attribution is standard; see [22, Thm. 6.16(6)]. The version for merely stable ideals is obtained in [1], §2, where the Koszul cycles exhibited for a stable ideal are observed to have pairwise vanishing products. None of this affects [9], whose Proposition 2.6 proves I_n strongly stable, so that both hypotheses hold; we note it only because the distinction matters if the argument is reused. An alternative route to the same conclusion runs through componentwise linearity [19], since stable ideals have linear quotients.

5. Conjecture 4.6

By Corollary 4.4 of [9], the Poincaré–Betti series of the residue field over $A_n = \Gamma_n$ is

$$P_K^{A_n}(t) = \frac{(1+t)^r}{D_n(t)}, \qquad D_n(t) = 1 - t^2 \sum_{j=1}^{r} (1+t)^{r-j} C_{n,j}, \tag{17}$$

with $r = \pi(n)$. Conjecture 4.6 of [9] asserted that this fraction reduces to

$$P_K^{A_n}(t) = -\frac{(1+t)^{\ell_1(n)}}{q_n(t)}, \qquad q_n(t) = \sum_{i=0}^{\ell_2(n)} h_i(n) t^i, \tag{18}$$

with (1) $q_n(-1) \neq 0$; (2) $\ell_1(n) = \#\{p \text{ odd prime} \mid p^2 \le n\}$; (3) $\ell_2(n) = \ell_1(n) + 1$; (4) $h_0(n) = -1$; (5) $h_1(n) = \pi(n) - \ell_1(n)$; (6) $h_{\ell_2(n)}(n) = C_{n,1} = \lceil n/2 \rceil$. We prove all six.

The point of the next lemma is that the whole conjecture is controlled by a single quantity: the order of vanishing of D_n at $t = -1$.

Lemma 13. *Put $u = 1 + t$ and $c_k = C_{n,r-k}$ for $0 \le k \le r-1$, with $c_k = 0$ otherwise. Then the coefficient of u^0 in D_n is $1 - c_0$, and for $k \ge 1$ the coefficient of u^k is $-(c_k - 2c_{k-1} + c_{k-2})$.*

Proof. Reindexing (17) by $k = r - j$ gives $D_n = 1 - (u-1)^2 T(u)$ with $T(u) = \sum_{k=0}^{r-1} c_k u^k$; expand $(u-1)^2 = u^2 - 2u + 1$. □

Lemma 14. *$c_0 = C_{n,\pi(n)} = 1$ for every $n \geq 2$. Consequently $D_n(-1) = 0$.*

Proof. $\Phi(n, p_r)$ counts the integers in $[1, n]$ with no prime factor $\leq p_r$; as p_r is the largest prime $\leq n$, the only such integer is 1. Now apply Theorem 4 and Lemma 13. □

Proposition 15. *The order of vanishing of $D_n(t)$ at $t = -1$ equals $\pi(n) - \ell_1(n)$.*

Proof. By Lemmas 13 and 14 the order is the least $k \geq 1$ with $c_k - 2c_{k-1} + c_{k-2} \neq 0$. Since $c_{-1} = 0$ and $c_0 = 1$, an induction shows that the second differences vanish for all $k \leq K$ precisely when $c_k = k+1$ for all $k \leq K$; so the order is the least $k \geq 1$ with $c_k \neq k+1$, that is, with $C_{n,r-k} \neq r-(r-k)+1$.

By Corollary 6, $C_{n,v} \neq r - v + 1$ if and only if $p_{v+1}^2 \leq n$, i.e. $p_{v+1} \leq \sqrt{n}$. The largest v with this property is $v = \pi(\lfloor\sqrt{n}\rfloor) - 1$, and for $n \geq 4$ this equals $\#\{p \text{ odd} \mid p^2 \leq n\} = \ell_1(n)$, the two counts differing only by the prime 2. Hence the least admissible k is $r - \ell_1(n)$. (If no such v exists, i.e. $\ell_1(n) = 0$, then $c_k = k + 1$ for all $k \leq r - 1$ and the first non-vanishing coefficient is that of u^r, namely $-(0 - 2r + (r-1)) = r + 1 \neq 0$; the formula $r - \ell_1(n) = r$ still holds.) □

Theorem 16. *Conjecture 4.6 of* [9] *holds: with $\ell_1(n)$ and $\ell_2(n)$ as above,* (18) *holds together with clauses (1)–(6).*

Proof. By Proposition 15 we may write $D_n(t) = (1+t)^{r-\ell_1} d(t)$ with $d(-1) \neq 0$; set $q_n = -d$. Then (17) gives $P_K^{A_n} = (1+t)^r / \big((1+t)^{r-\ell_1}(-q_n)\big) = -(1+t)^{\ell_1}/q_n$, which is (18) together with clause (2). Clause (1) is $d(-1) \neq 0$.

For the remaining clauses, note first that $D_n(0) = 1$ and $D_n'(0) = 0$, since $D_n - 1 = -t^2(\cdots)$ vanishes to order 2 at $t = 0$; and that $\deg D_n = r + 1$ with leading coefficient $-C_{n,1}$, the top-degree term of (17) arising only from $j = 1$.

(3) We have

$$\deg q_n = \deg D_n - (r - \ell_1) = (r+1) - (r - \ell_1) = \ell_1 + 1 = \ell_2.$$

(4) $h_0 = q_n(0) = -D_n(0)/1^{r-\ell_1} = -1$.

(5) Differentiating $q_n = -D_n(1+t)^{-(r-\ell_1)}$ at $t = 0$ gives

$$h_1 = q_n'(0) = -D_n'(0) + (r - \ell_1) D_n(0) = r - \ell_1.$$

(6) The leading coefficient of q_n is minus that of D_n, namely $C_{n,1}$, which is $\Phi(n, 2) = \lceil n/2 \rceil$ by Theorem 4. □

Remark 17. Clause (2) — the identification of the numerator exponent — was the whole difficulty, and Proposition 15 shows why: it is the assertion that the sequence $v \mapsto C_{n,v}$ is exactly linear, of slope -1, on the range $v > \ell_1(n)$, and departs from linearity at $v = \ell_1(n)$. Corollary 6 makes both halves of that assertion visible at once, because it measures the departure from linearity by $E(n, v)$, a count of composite rough numbers which is positive exactly when $p_{v+1}^2 \leq n$. Figure 1 shows the effect.

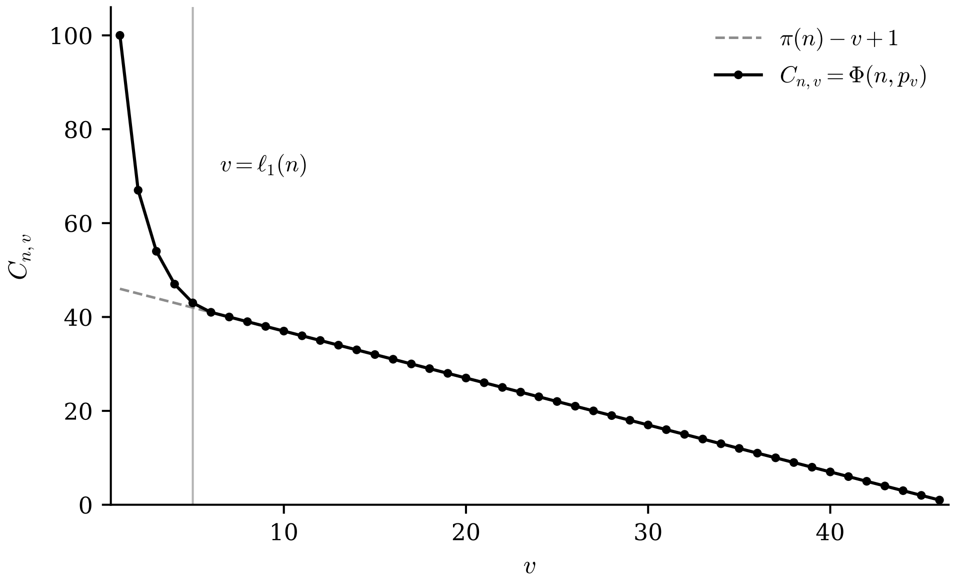

FIGURE 1. The profile $v \mapsto C_{n,v} = \Phi(n, p_v)$ for $n = 200$, against the line $\pi(n) - v + 1$ of Corollary 6. The vertical distance between them is $E(n, v)$, the number of composite p_v-rough integers $\leq n$; here $\pi(n) = 46$ and $\ell_1(n) = 5$, and the two coincide for every $v > 5$. That the profile is exactly linear of slope -1 beyond $v = \ell_1(n)$, and only there, is what Proposition 15 extracts as the order of vanishing of D_n at $t = -1$.

6. The average order of C_n

Theorem 3.4(3) of [9] gives the lower bound $C_n \geq \binom{\pi(n)+1}{2}$, proved there by exhibiting one minimal generator for each support of size 1 or 2. We show that essentially nothing else contributes in the limit.

Theorem 18. $0 \leq C_n - \binom{\pi(n)+1}{2} \leq n\big(\pi(\sqrt{n}) - 1\big) = O\Big(n^{3/2}/\log n\Big)$, *and consequently*

$$C_n \sim \binom{\pi(n)+1}{2} \sim \frac{\pi(n)^2}{2} \sim \frac{n^2}{2\log^2 n}.$$

In particular the bound of Theorem 3.4(3) of [9] *is asymptotically an equality.*

Proof. By Corollary 7 the difference equals $\sum_{x \leq n \text{ composite}}(\pi(\operatorname{lpf} x) - 1)$, a sum of non-negative terms. If $x \leq n$ is composite then $\operatorname{lpf} x \leq \sqrt{x} \leq \sqrt{n}$, so each term is at most $\pi(\sqrt{n}) - 1$; there are fewer than n composites $\leq n$. The prime number theorem gives $\pi(\sqrt{n}) \sim 2\sqrt{n}/\log n$, whence the stated O-bound, and $\binom{r+1}{2} \sim \pi(n)^2/2 \sim n^2/(2\log^2 n)$, which dominates it. $\square$

Corollary 19. $C_n =$ *A182843*$(n+1)$.

Proof. A182843(m) is defined as the number of composite integers $\geq m$ all of whose proper divisors are less than m. A composite W has largest proper divisor $W/\operatorname{lpf} W$, so the defining condition for $m = n+1$ reads $W > n$ and $W \le n \operatorname{lpf} W$; these are exactly the W corresponding to minimal generators of I_n, a monomial of weight W lying in $G(I_n)$ precisely when $W > n$ and $W/p \le n$ for every prime $p \mid W$, the binding case being $p = \operatorname{lpf} W$. □

Remark 20. The upper bound in Theorem 18 is far from best possible: almost all integers are even and contribute nothing. Writing the error term according to $\operatorname{lpf} x = p_j$ turns it into a Mertens-type sum

$$C_n - \binom{\pi(n)+1}{2} = \sum_{j\ge 2} (j-1)\,\#\{\, x \le n \mid x \text{ composite},\ \operatorname{lpf} x = p_j \,\},$$

and Table 1 suggests its true order is about $n^{3/2}/\log^3 n$. Determining it exactly is a question of sieve theory rather than of algebra, and the author intends to return to it separately.

n	C_n	$\binom{\pi(n)+1}{2}$	ratio
10^2	362	325	1.1138
10^3	14 957	14 196	1.0536
10^4	769 090	755 835	1.0175
10^5	46 233 662	46 008 028	1.0049
10^6	3 084 943 710	3 081 007 251	1.0013
10^7	220 905 096 960	220 832 955 910	1.00033

TABLE 1. C_n against the lower bound of Theorem 3.4(3) of [9].

7. Related work

Two remarks on the surrounding literature, both of which bear on how the result above should be read.

First, on the algebraic side, the ring Γ itself continues to be studied — recently in [18], which shows it is neither Noetherian nor Artinian, constructs several families of prime ideals, and computes its Krull dimension to be infinite. Its finite-dimensional truncations appear not to have been taken up by anyone else, but they were taken up once more by the present author, from a different direction: $\{1, 2, \dots, n\}$ is a *shifted multicomplex*, the complement of a strongly stable ideal being to a multicomplex what a shifted simplicial complex is to a squarefree one, and [11] accordingly computes the spectrum of its Laplacian. By the theorem of Duval and Reiner [12] that spectrum is integral, and [11, Cor. 14] identifies its multiplicities as the arithmetic counts

$$t_{i,k}(n) = \#\{\, 1 < m \le n \mid \Omega(m) = k,\ p_i \mid \operatorname{sfp}(m) \,\},$$

sfp denoting the squarefree part. These are not the counts $C_{n,v}$ of the present note — they fix the total degree and impose a divisibility condition, rather than fixing the least support — so neither result subsumes the other, and it is a natural question, and the exact analogue of Theorem 4, whether $t_{i,k}$ too admits a sieve-theoretic closed form. Artinian truncations of Stanley–Reisner rings are an active topic [23], and Γ_n is an object of that general shape, but no arithmetic enters there.

Second, and more to the point, the bridge Theorem 4 builds appears to be new. The asymptotic behaviour of $\Phi(x, y)$ is classical: Buchstab [13] obtained $\Phi(x, x^{1/u}) \sim u\,\omega(u)\,x/\log x$ in terms of the function that now bears his name, and de Bruijn [14] gave an approximation uniform for all $x \geq y \geq 2$; see [24] for a textbook account, [16] for numerically explicit estimates, [17] for short intervals, and [20] for the regime of fixed p, which is the one closest to ours. We have not been able to find anywhere in the literature a connection between these counts and any invariant of a free resolution; the two subjects appear not to cite each other at all. (A search for "Buchstab" in commutative algebra is apt to return instead the bigraded Betti numbers of face rings studied by V. M. Buchstaber, which is an unrelated subject and an unrelated person.)

8. What is next

Theorem 4 moves the combinatorics of $G(I_n)$ wholesale into sieve theory, and the questions it raises are correspondingly analytic.

Open problem 21. Give asymptotics for the whole profile $v \mapsto C_{n,v} = \Phi(n, p_v)$, *uniformly* in v, and deduce asymptotics for the graded Betti numbers of Γ_n through Corollary 4.2 of [9]. The behaviour of $\Phi(x, y)$ is classical [13, 14, 24], but here $y = p_v$ sweeps the whole interval $[2, n]$ as v does, so what is wanted is a statement uniform across all of it, including the two degenerate ends $v = 1$, where $\Phi(n, 2) = \lceil n/2 \rceil$, and $v = \pi(n)$, where $\Phi(n, p_r) = 1$. The uniform approximation of [14] and the explicit estimates of [16] are the natural starting points.

Open problem 22. Theorem 4 counts minimal generators by least support only. Is there a comparable sieve-theoretic description of the refined counts $C_{n,v,d}$, which additionally fix the total degree $d = \Omega$? A natural guess is a Φ-analogue restricted to integers with exactly d prime factors; the tables in Figure 2 of [9] are the data to test it against.

Open problem 23. Dirichlet convolution is the coarsest of Narkiewicz's regular convolutions. Carry out the analysis of [9] and of this note for the unitary, ternary, and general greedy convolutions: which truncation ideals are stable, and does an analogue of Theorem 4 hold? For the unitary case the truncations were treated by Stanley–Reisner methods in [10], and it is not clear that a sifting function appears there at all. This is work in progress by the author, who intends to treat the unitary case next.

Open problem 24. The elementary half of this note involves no homological algebra and is within reach of a proof assistant. Part of it has been formalised in Lean 4 and Mathlib, and is available in [26]: Corollary 6, and the whole polynomial argument of Section 5 — Lemma 13, Proposition 15 and clauses (1), (2), (4), (5) of Theorem 16 — are proved outright, with no `sorry` and on the standard axioms. What remains is (i) clauses (3) and (6), which are degree and leading-coefficient computations, fiddly rather than deep; (ii) Propositions 10–11; and (iii) Theorems 3 and 4 themselves. Item (iii) is the real obstruction, and it is the same one that puts Section 5's homological input out of reach: the formalisation takes $C_{n,v} = \Phi(n, p_v)$ as a definition and checks it numerically, because $G(I_n)$ cannot even be stated without monomial ideals, which Mathlib does not have — any more than it has minimal free resolutions, graded Betti numbers or Golod rings.

Finally, three integer sequences arising here deserve to be in the OEIS. C_n is A182843, whose entry records none of the above; the sequence $C_{n,2}$ and the triangle $C_{n,v}$ appear not to be present at all.

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The whole subject of this note, and of [9] before it, is owed to **Johan Andersson**. It was he who suggested studying the truncations Γ_n in the first place, and he who pointed out that the ideals I_n are monomial — and that this ought, therefore, to be my cup of tea. It was, and the observation has now paid out twice.

Just as Geheimrat Goethe had the diligent Friedrich Wilhelm Riemer to help him write *Die Wahlverwandtschaften*, just as Grothendieck had Jean Dieudonné to help him complete EGA, I have Claude!

Concretely: the errata of Section 4 were found by Claude (Anthropic) while re-reading [9] with the author, by recomputing every table in that paper along a route independent of the one used there. Theorem 4 and its proof, the reduction of Conjecture 4.6 to the order of vanishing of D_n at $t = -1$ (Lemma 13 and Proposition 15), and the asymptotic Theorem 18 were likewise found by Claude, working with the author in an interactive session, as was the identification of C_n with A182843. Claude also located and verified the references, and drafted this manuscript, with the author directing the exposition. The author has checked the results to the best of his ability, made final edits, and assumes full responsibility for any errors, mistakes or gaps that remain.

It is a pleasure to acknowledge the computer algebra systems without which none of this would exist. The computations behind [9] were carried out in Maple: the generator counts of its Figures 1 and 2, and — assembled from those counts through the formulas of its Corollaries 4.2 and 4.4 — the Poincaré–Betti series of its Figure 3. In the companion papers on unitary convolution the division of labour was quite different. For [10], Maple was used to *generate* Macaulay2 [21] input — procedures emitting ring and ideal declarations line by line — and Macaulay2 then computed the free resolutions, Betti numbers and Poincaré–Betti series. The homology of the relevant simplicial complex was computed with the Simplicial Homology package [15] of Dumas, Heckenbach, Saunders and Welker for GAP [4]; having found it torsion-free, the author wrote a short GAP programme to test lex-shellability, which it duly was. Every computation in the present note was done in SageMath [8], and deliberately along a route different from the one taken in 1999 — the minimal generators of I_n are enumerated as integers rather than counted by the formulas under test — so that agreement between the two is evidence rather than tautology.

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Conflicts of interest

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